



Lecture 3: Dispatch and variable generation

Energy Economics of the Green Transition

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Today

A workshop on Hirth (2013): what happens to the value of wind as wind is built, and whether that is anyone's problem.

		min
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Breaks of fifteen minutes after part 1 and after part 2. The worksheet is on paper; the notebook is optional.

Intro

Energy System Models, sections 2 and 3; appendix A if the duality machinery is new.

Hirth (2013), *The market value of variable renewables*. Sections 1, 3, 4.3, 5.1–5.5 and 7 are the ones today's questions point at.

Optional: Joskow (2011) on why levelised cost is the wrong comparison; López Prol et al. (2020) for cannibalisation in market data.

The question

Hirth (2013): what is a MWh of wind worth, compared with the average MWh, and how does that change as wind is built?

His answer, three ways:

source	wind's value factor	solar's
literature review	1.1 near zero share, falling	falls faster
regression, 2001–2010 market data	−1.62 per point of share (thermal)	
European model, fixed fleet	0.5 at a 30% share	< 0.5 at 15%
European model, long run	0.65 at 30%	

And a claim: profile, balancing and grid-related costs “are not market failures”.

Today: **why** (part 1), **how much**, and **what drives it** (part 2), **so what** (part 3).

Setup: two hours, three plants

A calm hour and a windy hour of equal length, nothing linking them. Wind capacity K is the knob.

	capacity, GW	marginal cost, EUR/MWh	availability (calm, windy)
Wind	K	0	0, 1
Coal	2	40	1, 1
Gas	2	80	1, 1
Load, GW	calm 1.5, windy 3		

Predict before you compute: at a small K , is wind's value factor above or below one? Write your guess down, with the reason.

The value of wind in a two-hour system

1. The dispatch problem, its conditions, the price in each hour for every K .
2. $\bar{p}_w(K)$, $\bar{p}(K)$, $VF_w(K)$; sketch; name the kinks.
3. Hirth's time-weighted base price against the note's.
4. Short, mid or long term? $\partial C^*/\partial K$, and the capture price as a cost saving.
5. General T hours: monotonicity, and the sign of $VF_w - 1$ at zero share.

Stop point

$VF_w(K)$ with its kinks named, and the derivative of question 4.

The full questions and the stretch are on the worksheet.

Solutions 1

Wind capacity rises from 2 to 2.5 GW. The price in the windy hour

1. falls, because more zero-cost supply entered
2. is unchanged, because coal is still the interior plant
3. falls to zero, because wind is now marginal

Solutions 1: the numbers

Windy hour, residual load $3 - K$; the calm hour is coal at 40 throughout.

	interior plant	λ_2	VF_w
$K < 1$	gas	80	1.2
$1 < K < 3$	coal	40	1.0
$K > 3$	wind	0	0

Kinks at $K = 1$ (last MW of gas leaves) and $K = 3$ (last MW of coal leaves; curtailment starts).

Board: questions 4 and 5. K cancels in $\bar{p}_w = \sum_h \gamma_h \lambda_h / \sum_h \gamma_h$; one more MW walks λ_h down the staircase in the hours where $\gamma_h > 0$; the sign at zero share is $\sum_h (\gamma_h / \bar{\gamma} - D_h / \bar{D}) \lambda_h^0$.

Hirth's base at $K = 0.5$: $\langle \lambda \rangle = 60$, so 1.33 against 1.2. In general

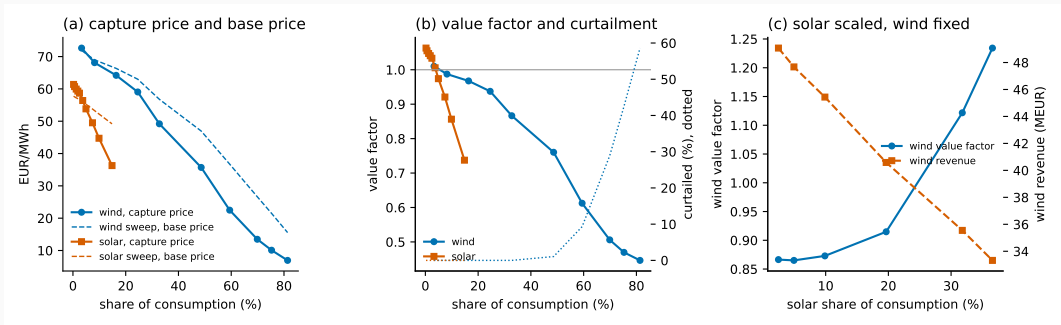
$$\bar{p} - \langle \lambda \rangle = \text{Cov}(D, \lambda) / \bar{D}.$$

Short term. Envelope:

$$\frac{\partial C^*}{\partial K} = - \sum_h \gamma_h \bar{\mu}_{w,h} = - \sum_h \gamma_h \lambda_h.$$

Part 2

Setup: the evidence



The note's hourly model on the ten-plant fleet of Table 1, DK1 2024 profiles, every eighth hour. Wind scaled; solar scaled; solar scaled with wind fixed. The table is on the worksheet; the notebook reproduces it, optionally.

The cannibalisation curve, and what drives it

1. Wind's slopes, 3–33% and 33–59%, against Hirth's regression; two reasons, with directions.
2. Falling without curtailment, then with it: the price-setting plant in each regime.
3. Solar's slope against wind's; the profile statistic that explains it.
4. Solar up, wind fixed: $\Delta \ln VF_w = \Delta \ln \bar{p}_w - \Delta \ln \bar{p}$; absolute and relative.

Stop point

Two slopes for wind, one ratio for solar, the two terms of the decomposition.

The full questions and the stretch are on the worksheet.

Solutions 2

Solutions 2: the curve against Hirth's numbers

Slope of the value factor, points per percentage point of share:

wind, 3–33%	−0.47
wind, 33–59%	−1.0
solar, 0–15%	−2.2
Hirth, thermal systems	−1.62
Hirth, hydro systems	−0.26

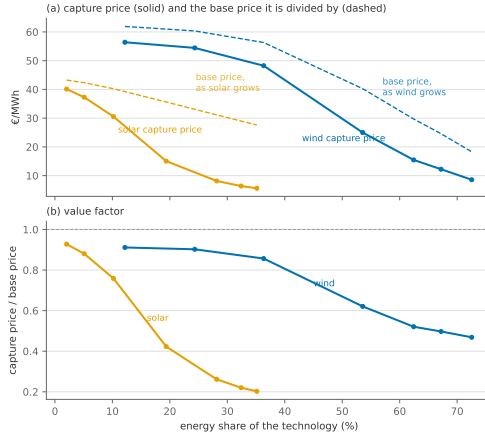
Solar up, wind fixed, $s = 1 \rightarrow 24$:

	$s = 1$	$s = 24$	$\Delta \ln$
$\bar{p}_w$	49.2	43.5	−0.12
$\bar{p}$	56.8	35.3	−0.48
VF_w	0.87	1.23	+0.35
wind revenue, MEUR	49	33	−0.39

The ratio rose because its denominator fell faster.

Two regimes on the figure: merit-order steps until $s \approx 1.5$, then wind's own step at zero with curtailment. Solar: a third of the hours, all the same hours.

The same experiment on the real DK1 fleet, with imports (note, figure 3.5)



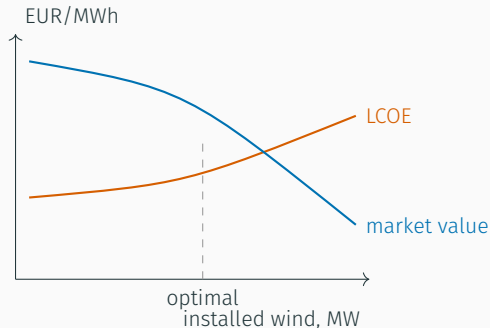
What differs from Hirth's Europe, and which way it pushes

difference	wind's value factor	where it returns
no exports: surplus is curtailed	lower	§3.6 bias box; lecture 5
no storage, no flexible demand	lower	lecture 4
flat merit order, nothing below coal	higher	Table 1
Danish wind correlated with Danish load	higher at low shares	Table 2
fleet fixed while wind grows	lower than long run	lecture 6

Hirth's own drivers (his Table 8) all act through the base price level and the slope of the merit order. The carbon price is his ambiguous case, and here it raises wind's value factor: $0.61 \rightarrow 0.74$ at $s = 2$.

Part 3

Setup: Hirth's figure 3, and his claim



Rising cost, falling value, and the intersection as the optimal amount.

“Profile, balancing, and grid-related costs are not market failures, but represent the intrinsic lower value of electricity during times of high supply, at remote sites, and the economic costs of uncertainty.”

Wind costs c^l per MW and year; $\Gamma = \sum_h \gamma_h$.

Optimal deployment, and whether the drop is a market failure

1. Hirth's figure from the model: free entry, the planner's condition, a subsidy σ , the loss.
2. The developer's argument: which of Hirth's three costs, market failure or not; \$3.5 against his sentence.
3. Joskow: what is the good, why is a MWh not a MWh; $K = 0.5$ and $K = 3.5$.

Stop point

The free-entry condition, and one paragraph placing the developer's argument on a named result.

The full questions and the stretch are on the worksheet.

Solutions 3

Solutions 3: the intersection is the optimum, and the drop is pecuniary

Free entry stops where

$$\Gamma \bar{p}_w(K) = c^l \iff \bar{p}_w(K) = LCOE_w,$$

and the planner sets $-\partial C^*/\partial K = c^l$, which by part 1 is the same equation (Result 6). A subsidy σ moves entry to $\bar{p}_w + \sigma = LCOE_w$; the loss is the area between the two curves right of the intersection.

Profile cost is the developer's argument: pecuniary, the price doing its job (Result 1). Not a market failure, and compensating it overshoots.

Grid and balancing are real externalities *when* unpriced: congestion under flat or zonal tariffs, reserve socialised. Nodal pricing; imbalance charges. One instrument per externality.

Joskow: the good is electricity in hour h . Wind's MWh is worth 80 at $K = 0.5$ and 0 at $K = 3.5$; coal's 57 and 40.

Wrap-up

What carries forward

1. **The price is a staircase in residual load.** Wind walks it down in the hours it produces, and is paid in exactly those hours. Result 1, hour by hour; the capture price is the marginal cost saving.
2. **The slope of the merit order is the sufficient statistic.** Every driver Hirth tests works through it or through the price level.
3. **Cannibalisation is pecuniary.** What it measures is missing flexibility: nothing in today's model can move a MWh from a surplus hour to a tight one.

Lecture 4 adds the first thing that can: storage. The question becomes what a battery is worth, and to whom.

Before lecture 4: *Energy System Models*, section 4. [\[paper\]](#). Full solutions to today's worksheet go up [\[when\]](#).

- Hirth, Lion (2013). **“The Market Value of Variable Renewables: The Effect of Solar Wind Power Variability on Their Relative Price”**. *Energy Economics* 38, pp. 218–236.
- Joskow, Paul L. (2011). **“Comparing the Costs of Intermittent and Dispatchable Electricity Generating Technologies”**. *American Economic Review: Papers & Proceedings* 101.3, pp. 238–241.
- López Prol, Javier, Karl W. Steininger, and David Zilberman (2020). **“The Cannibalization Effect of Wind and Solar in the California Wholesale Electricity Market”**. *Energy Economics* 85, p. 104552.